

The Normal Distribution

Introduction

The normal distribution, also known as the Gaussian distribution, is one of the most important probability distributions in statistics and natural sciences. It is characterized by its bell-shaped curve and is defined by two parameters: the mean μ and the standard deviation σ .

Mathematical Definition

The probability density function of a normal distribution is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where x is a random variable, μ is the mean, and σ is the standard deviation.

Properties

The normal distribution has several important properties. Approximately 68% of the data falls within one standard deviation of the mean, 95% within two standard deviations, and 99.7% within three standard deviations. This is known as the empirical rule or the 68-95-99.7 rule.

Many natural phenomena, such as heights and test scores, follow a normal distribution. Due to the Central Limit Theorem, the normal distribution is fundamental to statistical inference and hypothesis testing.